Boundary Conditions for the Digital Forensic Use of Electronic Evidence and The Need for Forensic Counter-Analysis

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Abstract— **Network and Digital Forensics provide information about electronic activity in new, sometimes unprecedented forms. These new forms offer new, powerful tactical tools for investigations of electronic malfeasance when incorporated under traditional legal regulation of state power, particular that of Fourth Amendment limitations on police searches and seizures under the U.S. Constitution. These tactical tools raise issues of public policy and privacy that may raise concerns about the proper police oversight of civil society. How those issues are resolved will define personal privacy, autonomy and dignity in the 21st digital century.**

***Index Terms*— digital, network, forensics, probably cause, fourth amendment, privacy, autonomy, dignity**

I. Introduction

THE tools of Network and Digital Forensics provide information about electronic activity in new, sometimes unprecedented forms. These new forms offer *new, powerful tactical tools* for investigations of electronic malfeasance when incorporated under traditional legal regulation of state power, particular that of Fourth Amendment limitations on police searches and seizures under the U.S. Constitution.

But the Fourth Amendment raises a variety of issues that relate to personal privacy, particularly those of personal autonomy and human dignity. Digital technologies change the ways people interact so as to challenge our notions of privacy and dignity. How the law responds to this may impact how we view privacy and the concept of personal dignity.

The "token" that generally permits state agents – the police – to invade the privacy and home life of citizens is the judicially-ordered "search warrant," issued by a magistrate upon a finding that there is "probable cause" to believe a crime or evidence of a crime may be found at a particular place or on a particular person.

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Absent this permission a person may not be otherwise forced to allow entry of state agents into their private personal affairs against their will.

The electronic realm does change what is factually private, something some may find difficult to accept. Network and cloud data offer profiles of individuals they may not expect.

Network forensics and electronic evidence were key to securing search warrants in many computer crime cases, such as the Innocent Images/Operation Candyman prosecutions for Internet child pornography. [1], [2], "The Case of the WiFi Spoofer" Internet threats investigation and the Bach seduction prosecution.[3], [4]

A series of court cases in the United States indicate how powerful tactical use of this data at the boundaries of the Fourth Amendment can be, even as courts rely on system- collected data independent of network forensic data of actual network activity. Tactical use of network forensic and system file data drives an investigation through "probable cause" for the search or seizure of computer systems that themselves prove the case. In such tactical use this data need only meet a lesser standard of reliability and weight that may alone be insufficient to show guilt or innocence.

Though effective, these tools raise issues of law, public policy and privacy as to proper police oversight of civil society. There is concern that existing rules fail to properly deal with digital evidence.[5] ""In the old days, the laws against illegal search and seizure were interpreted much more strictly," [defense counsel] Aldridge says, "but as this technology develops, the definition of probable cause will most likely be expanded." Id.

As noted by the dissent in *United States v. Gourde,* 440 F.3d 1065 (9th Cir.2006):

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> [in using computer evidence] … it is important that courts not grow lax in their duty to protect our right to privacy and that they remain vigilant against efforts to weaken our Fourth Amendment protections..

One commentator has poised the questions "Is the Fourth Amendment Relevant in the Technological Age?" [10]

II. Probable Cause of A Compute Crime That Justifies A State Invasion of Personal Privacy

The Construction of "Probable Cause" in American Law

Digital forensics investigations seek electronic evidence that either alone or when matched with other evidence shows a crime. It can be vital to correlate the electronic record with other actions.

This correlation and development role is particularly important for remote data collected over networks; correlation to other evidence is a key function of electronic evidence in prosecuting a digital crime. [6]

Legal state searches and seizures are essential for the collection of more, more strongly linked evidence of crime. But the fruits of an illegal search, once tainted, may be banned

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from use in criminal proceedings and the police authorities punished for their misconduct; standards for proper, legal searches are of great importance to police authorities.[7]

Those standards are based on the legal rules set by states to regulate police conduct. State power to investigate crime and malfeasance is controlled under the rule of law in most countries. Investigations of computer and network crimes fall under this regulation.

In the United States, the Fourth Amendment to the Constitution of the United States limits the power of police to search and seize a person, his computer and related transaction/content data:

> The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no warrants shall issue, but upon probable cause, supported by oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized. IV Amend. (1791)

Absent special circumstances, the search or seizure of a person or his effects without consent is illegal in the U.S. unless

1) an application under oath is made
2) before a neutral magistrate that
3) sets out facts to establish "probable cause" to believe a crime has been committed and evidence of that crime will be found in the place searched and things seized.

"Probable cause" itself means a "fair probability" under a commonsense analysis that evidence is to be found at the place to be searched; this was defined as being less that the "preponderance of the evidence" standard to support other judicial findings. *Illinois v. Gates*, 462 U.S. 213, 230, 246 (1983). An application for a search is to be judged under the "totality of the circumstances" presented. *Id.*

With the accumulation of network forensic and system data in so many forms, as well as the volatility and multiplicity of that data, what data is sufficient to say there is a fair probability that a particular network or system user has contraband or evidence of a crime on his or her computer? Once that quantum is defined, police power is essentially unlimited once that measure is met.

This sensitivity to constitutional principles is most strenuously tested when looking at the "heroin of cyberspace," child pornography. [8] It is one of the most inflammatory misuses of networks, yet it offers clarity of analysis precisely because it is a crime to have it in "digital possession."[9]

The Boundaries of "Fair Probability" of Digital Evidence of Crime and *United States v. Gourde,* 440 F.3d 1065 (9th Cir. 2006)

The boundaries of "Fair Probability" of the existence of criminal evidence are pushed by the decision of the U.S. Court of Appeals for the Ninth Circuit in *United States v. Gourde,* 440 F.3d 1065 (9th Cir. 2006). In that case the appeals court held, in essence, that an individual's on-line membership, as evidenced by remote system data, in a website with substantial, overt child pornography was sufficient to justify the search and seizure of that person's computer. This follows on decisions in *United States v. Martin*, 426 F.3d 68, 75 (2d Cir. 2005) ("It is common sense that an individual who joins such a site would more than likely download and possess such material.") & 426 F.3d. 83 (2nd Cir. 2005) (denying rehearing) and *United States v. Froman*, 355 F.3d 882, 890-91 (5th Cir. 2002) ("It is common sense that a person who voluntarily joins a group such as Candyman, remains a member of the group for …a month without cancelling … and uses screen names that reflect his interest in child pornography, would download such pornography from the website and have it in his possession.") supporting such a search and seizure.

But in determining that a "fair probability" existed that Gourde had downloaded child pornography to his computer, there was no evidence of network activity transferring contraband files to Gourde's computer, either through express download/ftp transfer,e-mail or simple http transfer via the web browser.

This case sparked concern over the boundaries of network-transmitted data and user liability. [10] echoing concerns raised by Operation Candyman that ""One click, you're guilty," says an FBI agent."A federal offense is that easy."" [11]

General Principle or Fact Specific – The Affidavit for the Search

Micah Gourde pled guilty to possession of 100 computer images of child pornography (18 U.S.C. §§ 2252 and 2252A) but reserved the right to challenge the FBI's seizure of his computer, where the definitive evidence of his crime was found. If Gourde showed there was no probable cause to believe there was evidence of a crime on his computer, the seizure and search would have been illegal and the evidence found could not be used against him; as there was no other evidence, his conviction would not stand and he would be released.

What distinguished Gourde's case from others was that there was no direct evidence of possession of these illegal images by Gourde.

Rather, the facts stated by the FBI in the Moriguchi affidavit in support for the search warrant were , in essence, that

1) Lolitagurls was a subscription child pornography website,
2) Gourde's "steps to affirmatively join" the website, were shown by membership data, which included his credit card,
3) The "membership" was not cancelled,
4) any time someone visited the website, he had to have seen images of "naked prepubescent females with a caption that described them as twelve to seventeen-year-old girls."
5) Child pornography users collect and associate with other like-minded.

What was not raised in the affidavit was any evidence that Gourde had actually downloaded child pornography images or that it was Gourde himself that had joined the website.

This is similar to *United States v. Martin*, the supporting affidavit was deemed sufficient where it showed

> …that the overriding, if not the sole, purpose of the girls12-16 e-group was illicit (to facilitate the receipt and distribution of child pornography); *that an e-mail address of a girls12-16 member was linked to Martin's house;* that collectors of child pornography overwhelmingly use the internet and computers to distribute and hoard this illegal pornographic material; and that, accordingly, there was a "fair probability," given the totality of the circumstances and common sense, Gates, 462 U.S. at 238, that evidence of a crime would be found at Martin's home because membership in the e-group reasonably implied use of the website." (emphasis added) 426 F.3d at 74.

This was the same rationale seen in *United States v. Froman*, where the supporting affidavit set out the following facts or inferences:

> 1. Froman joined Candyman on January 9, 2001, and remained a member until it was shut down on February 6, 2001.

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2. The *sole* purpose of the Candyman eGroup, as demonstrated by the statement in its website and the activities generated on the website during the time Agent Binney was a member, was to receive and distribute child pornography and erotica. All members were given instructions for cancelling membership in the group, and Froman at no time cancelled his membership.
3. The website enabled members to upload and download images of child pornography, and members even provided input as to the types of images they preferred. Agent Binney described the hundreds of images of child pornography he captured or received in e-mails through his membership in the group. Members were also directed to web pages with similar illegal content.
4. Froman registered a number of screen names with AOL that reflected his interest in child pornography, including Littlebuttsue and Littletitgirly

Minimal evidence supports the issuance of a state warrant to search and seize an implicated computer system.

The *Martin* rationale was criticized by other judges of that circuit in *United States v. Coreas*, 419 F.3d 151 (2d Cir. 2005) where the court found, given the falsehood in the affidavit, that the fruits of a poisoned search should be set aside. This left the only evidence of Coreas' involvement with the website that he clicked on a button in an email that automatically made him a website member. Although the court felt this was insufficient and the evidence should be suppressed, it felt based on the earlier 2nd Circuit precedent of *Martin* and "we believe Martin itself was wrongly decided and sets a "dangerous precedent." See *Martin*, slip op. at 22 (Pooler, J., dissenting). Nonetheless, since the Martin case was heard first, we are compelled, under established rules of this circuit, to affirm Coreas' conviction." Similarly, In *United States v. Coreas*, 259 F. Supp. 2d 218 (E.D.N.Y. 2003) the district court explains its rational that child pornography user characteristics and membership is enough.

The *Martin* court, in its opinion denying Martin a rehearing and responding to criticism in *United States v. Coreas* and from its own Judge Poole said

> The internet is not a safe haven for illicit conduct. Rather, it is a digital community where the zeros and ones that translate into visible and audible expressions have legal consequences. Although we will be diligent to guard against unlawful searches and seizures, even at the digital divide, the internet does not present an exception to established principles of probable cause. *Id*., at 89

III IMPLICATIONS WITH NETWORK FORENSICS AND OTHER SYSTEM DATA

These cases imply an exceptionally low standard of electronic evidence in support of the power of the search warrant. In some instances, the power of digital forensics to locate, acquire and authenticate the electronic evidence is bypassed once a path to potential evidence is described.

The ease of fabrication of electronic evidence goes far beyond what is possible with other media and is accomplished through common, non-technical means. [8] This creates a possibility for exceptional abuse through the application of police power.

Cyberextortion using child pornography is an issue world wide.[12], [13, 14] Such standards offer opportunities to abuse through planted and spoofed evidence without incentive to authenticate data or correlate it to other evidence of criminal activity.

On the other hand, the shear poison of child pornography may lead the courts and justice agencies to treat it differently than other criminal activity such that this seemingly lesser standard does not apply elsewhere. The ethical overflow is Operation Candyman, where search warrants for on-line child pornography were secured on the basis of an FBI agent's affidavit that contained false statements about network and website operations and how visitors to the Candyman website interacted with that site. Examples include the district court cases *United States v. Strauser,* 247 F. Supp. 2d 1135, 1142 (E.D. Mo. 2003) and *United States v. Perez*, 247 F. Supp. 2d 459, 479-480 (S.D.N.Y. 2003). Yet when this material falsity was raised in *Martin, Foreman* and *Coreas,* above, the courts relied on system data to validate the police action.

Judge Poole's dissent in *Martin* accuses the court of creating just such an exception; the danger, as may be inferred from the *Coreas* opinion, is that this exception might become the rule.

IV.TECHNICAL EXAMPLE: VISUAL CRYPTOGRAPHY

One example of how electronic evidence can be faked and an innocent party implicated is that from visual cryptography. Naor and Shamir at Eurocrypt '94 proposed the original visual cryptography method. [15] In its most basic form a black and white image is split into two shares (S1 and S2). [16], [17]

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S1 and S2 will have double the height and double the width of the original image.

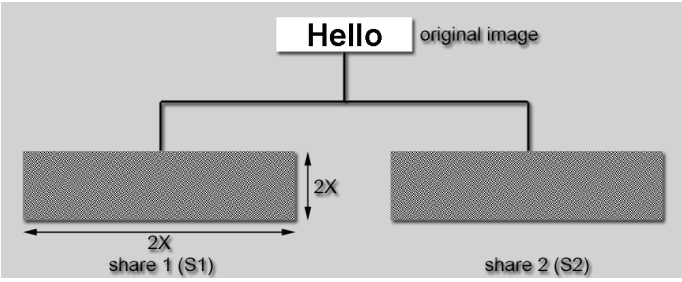

***Figure 1**: Generating two encrypted shares from an image*

Therefore, a single pixel in the original image is split into a set of four pixels. Any set of four pixels derived from the original pixel will alternate in color: black-white-white-black or white- black-black-white.

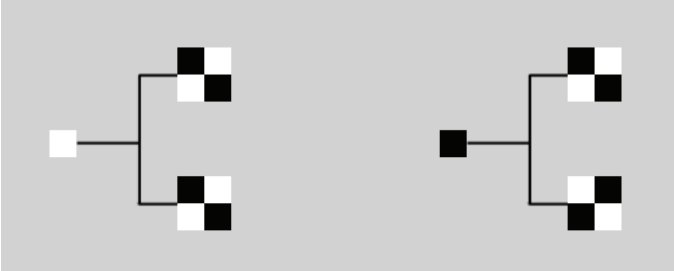

***Figure 2**: Possible sets of 4 pixels generated in share 1 and share 2 from encrypting a white or black pixel*

The specific order of the pixels in each four-pixel set in S1 is generated randomly. [18] The order in the complementary share 'S2' however, is not. It is generated based on the original pixel in the original image as well as the color and order of pixels in S1.

***Figure 3:** Two possible sets of 4 pixels generated in share 1 and share 2 from encrypting a black pixel*

So, if the original pixel was black, the two complementary pixels when superimposed should produce four subsequent black pixels. If, on the other hand, the original pixel was white, the superimposed pixels should still alternate producing a pseudo-grey color resulting from the alternating black and white pixels.

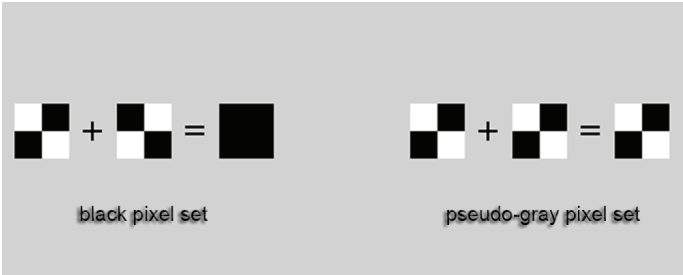

***Figure 4:** The result of superimposing corresponding sets of 4 pixels*

As a result, it is clear that this method is completely secure since the encryption is initially randomly generated. There is no way to tell whether a set of four pixels in an encrypted share is derived from a black or white pixel by just looking at one share.

This security, ironically, is the major Achilles' heel in this encryption method. If a single image is encrypted a number of times utilizing an algorithm which simply generates the encrypted pixels randomly every time the resultant shares will be different every single time as well. As a result any person who is in possession of a single share can neither confirm nor deny the absolute origin of that share based on that share alone. It is in this fact that an interpretation of a complementary image is possible utilizing any image desired as a base for the interpretation.

Visual Cryptography: Image Interpretation

The concept of generating a complementary share based on any image is possible due to the original encryption mechanism. In order to do that, the program goes through the pixels of the image that is to be generated one by one. It checks whether the pixel is black or white.

Secondly, it checks the corresponding set of four pixels in the base share. It checks the order, color and distribution of those four pixels.

Finally, based on the color of the original pixel and the order and color of the set of four pixels, another set of four pixels is generated in the complementary share. For example, in the retrieved share, a set of white-black-black-white can be complemented with black-white-white-black if the pixel we want to ultimately generate is black.

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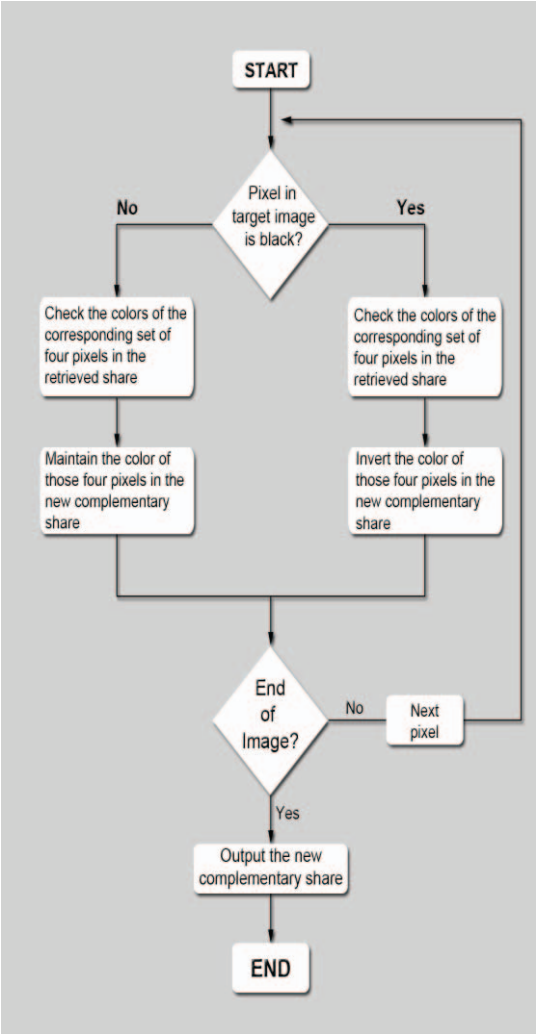

***Figure 5**: Complementary share generation flowchart*

As a result, one retrieved share can be used to generate a multitude of complementary shares that ultimately result in a multitude of interpreted images.

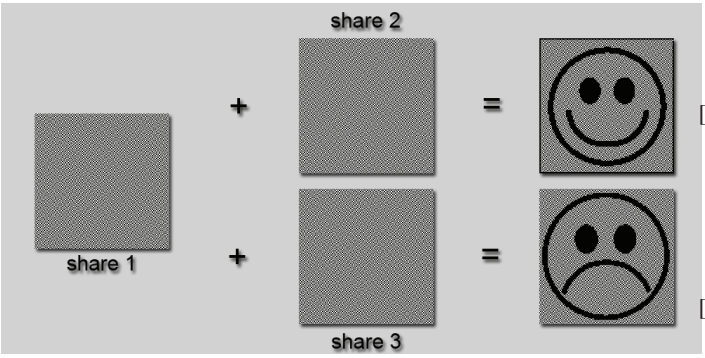

Figure 6: ***Two different images are generated using a single common share and two different complementary shares***

Thus data sufficient to currently support a search warrant against a private party may be spoofed. This is akin to the "

V.CONCLUSION

To the extent network and digital forensics seek the truth, not just convictions, this trend with electronic evidence may demand a forensics response. Rather to seeking to authenticate evidence of guilt, the discipline may need to assert the lack of authenticity and weight of guilt in some types of electronic evidence. That assertion may help balance where courts set the boundary for state searches and seizures based on electronic evidence of questionable reliability.

As the *Gourde* court said

> … Given the current environment of increasing government surveillance and the long memories of computers, we must not let the nature of the alleged crime, child pornography, skew our analysis or make us "lax" in our duty to guard the privacy protected by the Fourth Amendment. We are acutely aware that the digital universe poses particular challenges with respect to the Fourth Amendment.

That awareness of the law still needs knowledge of the facts of electronic evidence, its mutability and evanescence, to render good and just decisions.

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